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LITHIUM BATTERIES FOR PULSE POWER

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ABSTRACT

New designs of lithium batteries having bipolar construction and thin cell components possess the very low impedance that is necessary to deliver high-intensity current pulses. The R&D and understanding of the fundamental properties of these pulse batteries have reached an advanced level. Ranges of 50-300 kW/kg specific power and 80-130 Wh/kg specific energy have been demonstrated with experimental high-temperature lithium alloy/transition-metal disulfide rechargeable bipolar batteries in repeated 1- to 100-ms long pulses. Other versions are designed for repetitive power bursts that may last up to 20 or 30 s and yet may attain high specific power (1-10 kW/kg). Primary high-temperature Li-alloy/FeS₂ pulse batteries (thermal batteries) are already commercially available. Other high-temperature lithium systems may use chlorine or metal-oxide positive electrodes. Also under development are low-temperature pulse batteries: a 50-kW Li/SOCl₂ primary battery and an all solid-state, polymer-electrolyte secondary battery. Such pulse batteries could find use in commercial and military applications in the near future.

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1 INTRODUCTION

The pulse batteries considered in this paper are characterized by extremely high specific power (higher than 1 kW/kg) during pulses from milliseconds to several seconds. They also achieve moderate to high specific energy (40-150 Wh/kg, calculated as the sum of the energies of pulses in a discharge half-cycle). Among the pulse batteries under development, the lithium systems, especially the high-temperature versions, provide the best specific power and specific energy. Recent years have seen a fast pace in the research and development of promising pulse batteries. Pulse batteries could find use in commercial and military applications in the near future. This paper reviews the basic operational and design principles, as well as the representative versions of the lithium-based pulse battery systems.

Figure 1 shows the specific power and specific energy ranges of representative battery and capacitor systems. The dotted fields indicate the state-of-art battery and capacitor systems in a rather generalized manner. Location of a particular system in the indicated fields depends on the design and discharge parameters. Two general trends, nevertheless, can be observed. Namely, (1) the specific power and the available specific energy are in reciprocal relationship with each other, and (2) if longer cumulative pulse time is required either the specific energy or the specific power has to be sacrificed.

The hatched fields in Fig. 1 indicate new types of pulse-power systems under development, the

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electrochemical capacitors (ultracapacitors) and the super high power density bipolar batteries, which may be called galvanic "capacitors." The ultracapacitors, which show capacitor-like discharge characteristics, utilize charge accumulation in the electrochemical double layer of certain electrochemical cells. The charge and discharge of the ultracapacitors are not associated with a faradaic process (i.e., an electrochemical reaction). Therefore, the stored energy is relatively modest; however, may amount up to 5 Wh/kg. The ultracapacitors are also called electrochemical capacitors to distinguish them from electrolytic capacitors, such as the widely used aluminum- or tantalum-foil capacitors. In contrast to the ultracapacitors, the galvanic capacitors, such as the lithium systems discussed here, utilize electrochemical reactions; their charge/discharge mechanism relates in a certain extent that of the electrochemical capacitors. Their operation and design principles are discussed in Section 2 below. One may foresee that in the future newer and newer ultracapacitor, galvanic capacitor, and bipolar battery systems will join the families of the batteries and capacitors. So, the presently isolated system fields in Fig. 1 may join together to bridge the two extremes offering even more options to choose from for various applications.

2 OPERATIONAL AND DESIGN PRINCIPLES OF LITHIUM PULSE BATTERIES

The special operating conditions and design requirements of the high-power pulse batteries warrant a short review of the operational principles.

Low cell impedance and fast kinetics of the electrode active materials are the primary requirements to achieve ultra-high specific power from a battery. The fast kinetics is dictated by the combination of active material and electrolyte choices and is greatly enhanced at elevated temperatures. The lithium batteries, in general, possess good kinetics and among them the Li(-alloy)/molten salt/ MS_2 (where M is either Fe, Co, Ni, or a mixture of them) and the Li(-alloy)/molten salt/ Cl_2 systems, all of which operate at $>400^\circ\text{C}$, are especially well suited for pulse power application. On the other hand, cell resistance and impedance are primarily influenced by the design parameters of these batteries and only partly by the system chemistry. Furthermore, the magnitude of the cell impedance [1] and also the relative contribution of the resistive, capacitive, and inductive components [2] depend on the time requirements and power density of the pulse.

The pulse batteries designed for high specific power consist of serially connected bipolar cells. Bipolar cell construction with thin electrodes and separators is needed to minimize ohmic losses within the battery and facilitate high current density. The bipolar construction and uniform current density over the faces of the electrodes are important conditions; otherwise, high specific power ($>1 \text{ kW/kg}$) cannot be obtained [2]. Thin electrodes, separators, and bipolar plates have to be used to reduce the ohmic resistance. There is a trade-off, however, between electrode thickness and area capacity. In metal sulfide cells, a fraction of a millimeter can hold 40-80 mAh/cm². Consequently, the cell thickness is in the 0.1-0.2 cm range. The high power density and strict timing requirements of pulsing necessitate a very careful matching of the battery- and load-subsystem characteristics.

For performance analysis and cell design studies, a convenient parameter, which is easily measurable with small experimental cells, is the area-specific impedance (ASI_t , ohm·cm²). The area-specific impedance of a cell at any given pulse time (t) is calculated from the voltage loss measured at time t ($\Delta V_t = V_o - V_t$, where V_o is the open circuit cell voltage and V_t is the cell voltage at pulse time t) and the applied current density (i): $ASI_t = \Delta V_t / i$. This concept of ASI_t is strictly valid only under uniform current-density conditions of the electrodes. This criterion, which is generally valid for bipolar constructions, must be remembered when ASI_t is measured and used in modeling and scale-up calculations.

Two fundamental relationships among current density, voltage losses, and pulse time have been found for Li-alloy/metal sulfide cells [1]. First, the area-specific impedance is constant over a wide range of current densities during galvanostatic pulses. This relationship holds only if ASI_t is compared at identical pulse

times of different current densities. For example, Fig. 2 shows voltage losses measured at identical pulse times as a function of current density in high-temperature, bipolar Li-alloy/FeS₂ cells [1]. The straight lines indicate constant ASI_t over a rather wide range of current density. A similar relationship has been reported for other battery systems. One may generalize this relationship for those well-designed batteries in which electrode kinetics and mass-transport limitations are negligible and only ohmic limitations are significant. The second kind of relationship indicates that ASI_t increases logarithmically with pulse time t , as shown in Fig. 3.

Plots of ASI_t vs. pulse time and depth of discharge are characteristic for a cell or battery design [1]. These graphs provide valuable input data for modeling, scale-up, and cell design. Methods for modeling battery performance for various pulse modes have been elaborated for high-temperature Li-alloy/metal disulfide bipolar batteries, and the calculated results have been in excellent agreement with measured data [3, 4].

The advantage of active material couples that produce high cell voltage is shown by the following relationship between area-specific peak power (ASPP_t) and open-circuit cell voltage (V_o): $ASPP_t = V_o^2/(4ASI_t)$. Although this relationship suggests use of pure lithium as the negative active material, alloyed lithium is preferred in high-temperature cells designed for long mission times to prevent excessive self-discharge rates across the thin separators [5]. Self-discharge in these cells is caused by the lithium that is dissolved in the molten-salt electrolyte. The highly reactive dissolved lithium diffuses to the positive electrode, and by chemically reacting with the active material decreases the available cell capacity. Another advantage of the use of alloyed lithium is that certain compositions provide chemical overcharge protection and cell capacity equalization for secondary operation [6]. The chemical overcharge protection and the chemical means of cell capacity equalization are prerequisites for bipolar secondary battery constructions. It would be impractical to use voltage/current lead connections at every bipolar plate of the battery, although they can be applied in a battery of serially connected rectangular cells.

Choices of various positive active materials offer a wide range of cell voltages. Transition-metal disulfides (FeS₂, CoS₂, NiS₂, or mixtures of these sulfides) show very good electrode kinetics at elevated temperatures for high power density application. The open-circuit voltage of such disulfide cells is close to 2 V, depending on the composition of the Li-alloy negative electrode. Lithiated vanadium oxide and LiCoO₂ have been identified as other candidate active materials [7]. These oxides, in addition to greater open-circuit voltage, exhibit much better thermal stability than FeS₂, which decomposes rapidly above 550°C. A chlorine electrode on activated porous graphite combined with a Li-Al [8] or Li-Si [9] negative electrode has an open-circuit cell voltage of about 3.3 V and good kinetics. The above-listed positive active materials are used for high-power applications at high temperatures (400-550°C) in molten lithium-salt electrolytes (LiCl-KCl or LiF-LiCl-LiBr). Ambient-temperature lithium pulse batteries are also being developed. These are either specialized versions of the "conventional" lithium batteries such as the Li/SOCl₂ [10, 11] and the Li/TiS₂ [12] systems or new types, which use polymer electrolytes and active materials [13]. In most of the proposed applications of the pulse batteries not only the high power but the high pulse energy is the primary factor. Therefore, good electrode kinetics is very important to sustain long pulses without excessive power degradation.

Proper thermal management is another key requirement for high-power-density batteries. Extra care has to be taken because at high-current-density charge and discharge, a great amount of heat evolves in electrochemical cells as a result of the electrochemical reaction, electrode polarization, and I^2R heating. The heat production rate may approach the magnitude of the electrical power at close to peak-power loads. Pulsing mode and length, as well as the ratio of the open- and closed-circuit periods, obviously greatly influence the thermal management requirements. One of the advantages of the Li-alloy/transition-metal disulfide cells is the endothermic cell reaction in "upper-plateau" operation [14] (in this cycling mode, the cell is discharged to the monosulfide step only). The endothermic process helps the thermal management

and improves cell efficiency [15]. For the upper-plateau cells, proper cell designs make thermoneutral operation possible even at high current densities [3].

The high-performance secondary pulse batteries are in the early stage of development, therefore, their lifetime and the factors limiting it are not clearly known. In general, fast and deep discharge decreases cycle life. Nevertheless, more than 100 cycles were achieved with high-temperature metal-disulfide pulse batteries [4]. Based on this figure combined with the number of pulses available from a single discharge half-cycle, 10^6 short pulses can be projected for the full lifetime of a battery. Effective peripheral seal at the edge of the cells in a battery stack is very important to prevent electrolyte leakage and ensure long cycle life. Leaking electrolyte may interconnect electrolytically adjacent cells and produce parasitic currents, which ultimately destroy battery integrity. Primary lithium pulse batteries -- mainly the high-temperature Li-alloy/ FeS_2 batteries (thermal batteries) -- are more developed and have found applications because of the simpler requirements when only shelf-life has to be considered in addition to the discharge characteristics.

Thin cell designs have an inherent "built-in" safety factor. The thin electrodes of the cell contain a limited amount of active material, and the cells are separated from each other chemically by a bipolar plate. This construction limits violent chemical reaction during a massive short circuit in a cell or when severe mechanical damage occurs to the battery.

Batteries that are designed for extreme high-intensity pulsed power may be called "galvanic capacitors", although they are special batteries [1]. The galvanic capacitors combine the advantages of both the battery and the capacitor system: a high-energy flux is available (capacitor) at a relatively constant voltage (battery); the energy is stored within the cell in the form of the chemical energy at high energy density (battery) and is available at any time in a very flexible manner (capacitor); and the pulse length can be varied without the penalty of a fast-dropping voltage that is characteristic for capacitors. Also, because the energy source is the cell itself, unlike the case with capacitors, the pulses can be repeated many times without requiring a recharge, although after the capacity has been exhausted the secondary galvanic capacitor is rechargeable. The high-current-density pulses discharge the surface of the active material particles. This surface layer, however, becomes quickly charged from the inside of the particles during the rest periods between the pulses. In this sense, the galvanic capacitor is an intermediate system between the non-faradaic ultracapacitor and the conventional batteries, as suggested by Fig. 1. The extreme high power-to-energy ratio ($\text{W/kg}/(\text{Wh/kg})$), which has been demonstrated as up to $500\text{-}1000\text{ h}^{-1}$, is another factor that distinguishes the galvanic capacitors from conventional batteries. In this relationship, the specific power usually refers to the peak power at full charge for a specified pulse length.

3 REPRESENTATIVE VERSIONS

The extreme high current density capability of a galvanic system was demonstrated first with Li-alloy/transition-metal disulfide couples [1, 16]. These high-temperature cells [1] were built with very thin electrodes and separator plates; the total thickness of the cell was 0.1-0.2 cm, depending on the design. Li-Al/ FeS_2 and Li-Si/ CoS_2 rechargeable cells produced thousands of 1- to 10-ms long pulses from a single discharge at current densities of $10\text{-}23\text{ A/cm}^2$. Also demonstrated was the ability to sustain second-long pulses with only a little performance decrease. These cells were operated with LiF-LiCl-LiBr electrolyte at $490\text{-}495^\circ\text{C}$. Based on single-cell measurements, performance of 83 kW/kg and 144 Wh/kg was projected for bipolar cell stacks. Representative data (1-ms pulse) for three characteristic cell designs developed at Argonne National Laboratory (ANL) are reproduced in Table 1 from reference 1. Similar exploratory cell constructions from ANL have produced long pulses in the range of 10-20 s and still maintained very low area-specific impedance ($50\text{-}200\text{ mohm.cm}^2$ per cell).

Bipolar Li-Al/CoS₂ secondary batteries for the pulse power application are being developed at the Westinghouse Electric Corp., Naval Systems Division [3, 4, 17]. Based on data measured for small cells, the Li-alloy/LiF-LiCl-LiBr-MgO/CoS₂ system was optimized for two ranges of pulsed power: high power, short pulse time (100 ms) and high energy, long pulse time (4 s). Successful scale-up to 20-cm² area cells and to five-cell bipolar stacks indicates that this system is a very good candidate for full-size battery development. In 100-ms pulses these batteries achieved up to about 50 A/cm² current density and 300 kW/kg specific peak power; and the cumulative specific energy in discharge half-cycles was 111 Wh/kg [4]. Figure 4 shows characteristic data for a two-cell battery of 20-cm² area [4]. Up to 140 charge/discharge cycles were demonstrated. Intra-cell leakage was not detected in any of the tested bipolar stacks. Loss of integrity at the cell edge made of BN appears to be the failure mode. Development of bigger versions (20-cm² area stacks of 145 cells, as well as 200-cm² area single cells and batteries) is planned [5].

Technochem Co. has initiated a program to develop a Li-Al/LiCl-KCl/Cl₂ primary battery for the pulse power application [8]. This battery is based on a novel activation treatment for porous carbon and graphite materials for use as chlorine cathodes at atmospheric pressure. Experimental cells show very good power performance at temperatures of 450-475°C. In short pulses, area-specific power of 20-25 W/cm² has been demonstrated. The cell has yielded an average of 9 W/cm² power for 5-s duration pulses, and the cell performance is steady from pulse to pulse. Representative pulse duty cycles are shown in Fig. 5 [8]. Eagle-Picher Industries, Inc., is also experimenting with Li-Si/LiCl-KCl-MgO/Cl₂ primary pulse batteries [9].

The lithium pulse batteries are not limited to the molten-salt systems. A high-power bipolar Li/thionyl-chloride primary battery, which operates at close to ambient temperature, is under development at Yardney Technical Products, Inc. [10]. The battery is designed either for 50 kW power output for a 5-min discharge or for a minimum of 4.4 kW/kg specific power at short pulses over a 8-10 year lifetime. Hermetically sealed cells made with 1.6 M LiGaCl₄-SOCl₂ electrolyte can sustain discharge at 200 mA/cm² in 2-, 4-, and 20-ms pulses with minimal voltage delay. Cathodes are composed of a blend of platinized carbons: low-surface-area Shawinigan Black (60 m²/g), high-surface-area carbon (1000-1500 m²/g), and Teflon binder. The discharge characteristics for a four-cell stack of full-size cells (500 cm²) in 400-mA/cm² pulses of 1-s duration are shown in Fig. 6 [10]. Even though the high-rate configuration suggests a hazardous battery design, Yardney's safety record indicates that it can be operated safely when certain, not disclosed, design considerations are applied.

Another example of a low-temperature system is an all solid-state construction being developed at Harwell Laboratory, UK. This is a rechargeable Li/poly(ethylene-oxide)₁₂LiClO₄/V₆O₁₃ system operating at 120°C for pulse lengths of 10 ms to 20 s [13]. This polymer-electrolyte cell offers simple, rugged construction; its specific power, however, is modest compared to that of other above-described systems, but very impressive considering it is an all solid-state construction. For repetitive pulses in excess of 1-s duration, high initial currents (e.g., 1.5 A from a 40 mAh cell) up to about 50% cumulative utilization have been observed (Fig. 7). For pulses of 10 ms, the cells have exhibited a rapid response and recovery, high current capability, and long pulse life. Nine thousand pulses of 1.3 A as an average have been obtained from a 110 mAh cell with a cumulative cathode utilization of 25% (Fig. 8). The results of this work have demonstrated that, when operated in a pulse mode, the polymer-electrolyte lithium battery technology can attain high specific power. Based on data from small cells, about 1 kW/kg specific power and 100 W/kg specific energy were projected for large cells.

4 APPLICATIONS

Pulse batteries could have various military and commercial applications. Military applications such as power source for electromagnetic launchers were considered first. Later other pulse power systems

(particle-beam accelerators, lasers, pulsed radars, microwave generators, and metal-ion beams for ion implantation) were proposed. Li-alloy/FeS₂ primary batteries (thermal batteries) that operate at high temperature have already found military and other applications.

A new commercial application for the pulse power batteries could be with battery-battery hybrids for electric vehicle propulsion and power conditioning. In this application, the pulse battery or ultracapacitor would provide the most part of the high power needed for acceleration or hill climbing. The main battery (or a fuel cell), which would be designed for high specific energy, would serve as the general energy source for average power loads and for recharging the high-power pulse battery. Reports on battery-ultracapacitor hybrids [18, 19] have shown the feasibility of this concept. The galvanic capacitors, however, as proposed by ANL, would fit better to the battery-battery hybrid application because their specific energy is about 30 times higher than that of the ultracapacitor.

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Table 1. Data for ANL's Li-Alloy/MS₂ Bipolar Cells

Design parameters			
Positive Electrode	FeS ₂	CoS ₂	FeS ₂
Separator	MgO	BN+MgO	BN+MgO
Electrolyte	LiCl-KCl	LiF-LiCl-LiBr	LiF-LiCl-LiBr
Negative Electrode	Li-Si	Li-Si	Li-Al
Bipolar Plate	Mo	Mo	Mo
Mass, g/cm ²	0.392	0.343	0.292
Thickness, cm	0.202	0.152	0.123
Area Loading, mAh/cm ²	42	42	42
Performance parameters: ^a			
ASL _i , mohm-cm ²	105	175	41
ASPP _i , W/cm ²	8.84	4.85	24.1
Specific Energy, ^b Wh/kg	105	122	144
Energy Density, ^b Wh/l	204	276	341
Specific Power, kW/kg	22.1	14.1	82.7
Power Density, kW/l	42.9	31.8	196
Power to Energy, ^c W/Wh	210	116	574

^aPerformance parameters were measured at t = 1 ms pulse time, 10% depth of discharge, and 495 °C and are given for peak-power condition.

^bCumulative energy of 1-ms pulses available in one discharge half-cycle.

^cRatio of the instantaneous specific peak power at the end of a pulse to the specific energy in a discharge half-cycle.

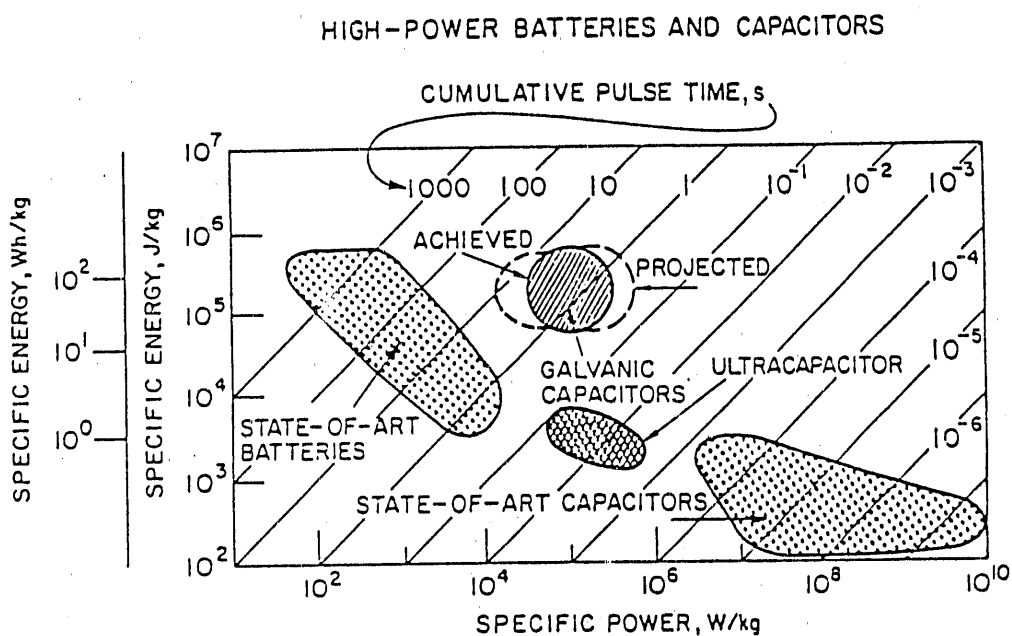


Figure 1. Performance fields of state-of-art batteries and capacitors, and the new galvanic capacitors and ultracapacitors (electrochemical capacitors). Description of the electrochemical capacitors can be found in references 20 and 21.

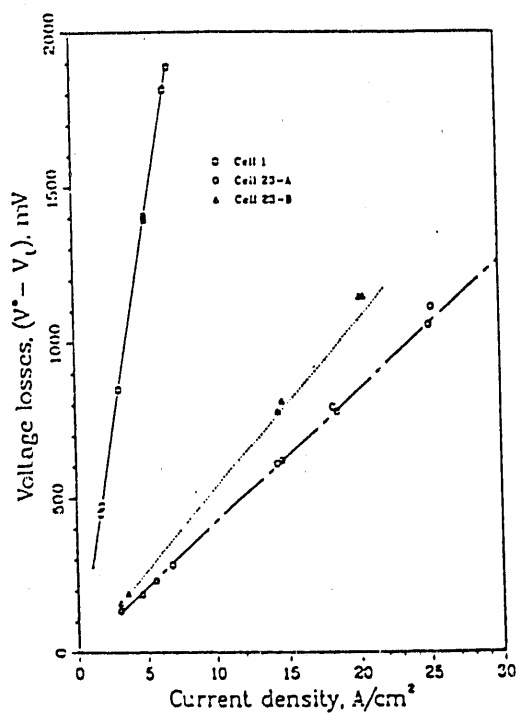


Figure 2. Voltage losses in two Li-alloy/transition-metal sulfide bipolar cells (1 and 23) compared at 1-ms pulse time and 490°C. A and B are measured at different depth of discharge of Cell 23 [1].

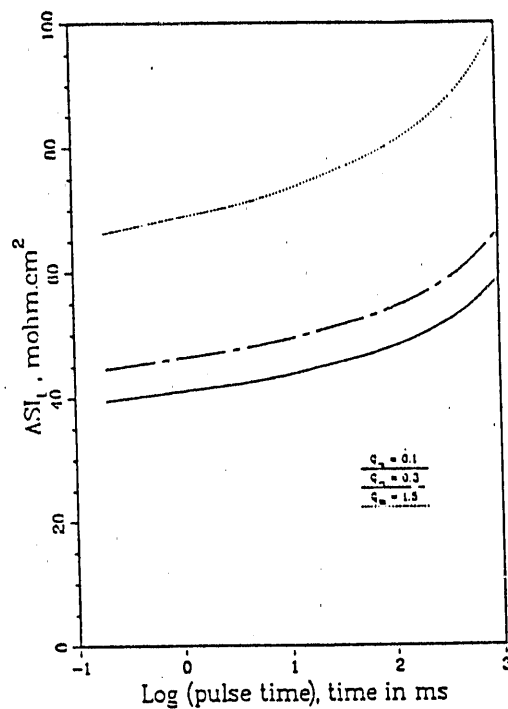
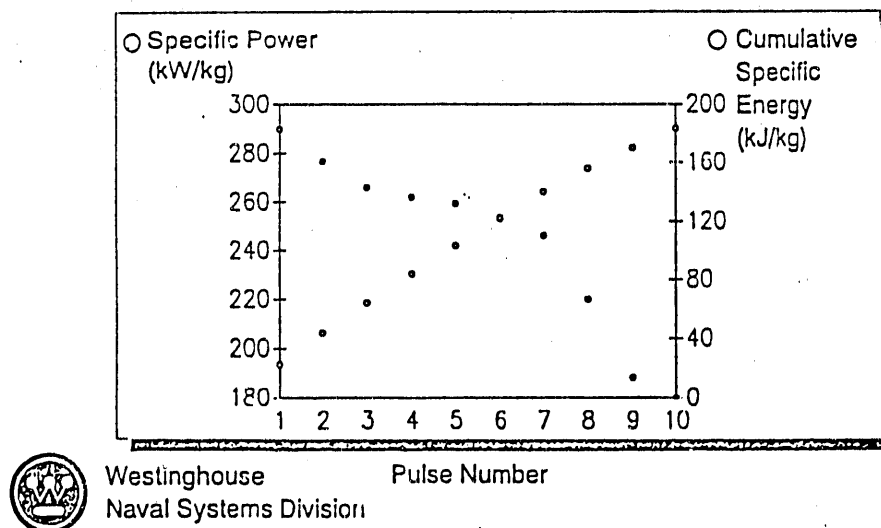


Figure 3. Area-specific impedance as function of pulse time measured in cell B at 490°C. ASL_i is compared at three depth of discharges: Q_m = 2 is equal to 100% depth of discharge [1].

20 cm² Two Cell Battery - Specific Power and Energy for a 10 (100 msec.) Pulse Cycle



Westinghouse
Naval Systems Division

Pulse Number

Figure 4. Performance data of a Li-Al/CoS₂ bipolar pulse battery measured at 480°C [4].

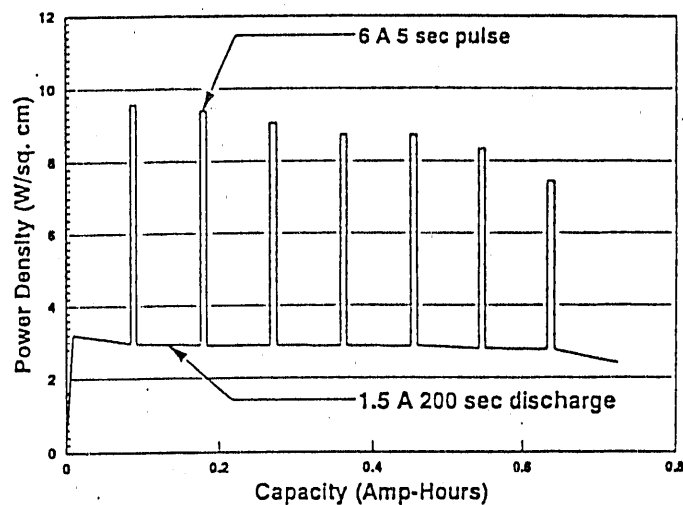


Figure 5. Power density as function of delivered capacity for a Li-Al/Cl₂ cell at 450°C [8]; electrode area is 1.27 cm².

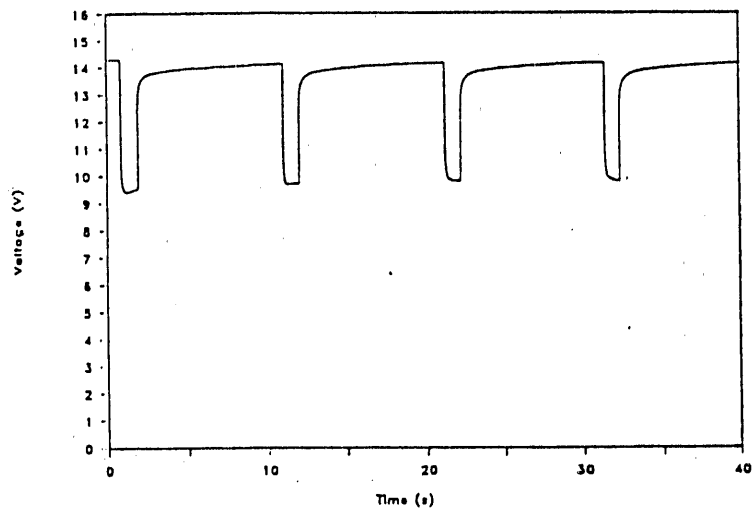


Figure 6. Pulsing characteristics of a Li/SOCl₂ multicell bipolar stack; 10% pulse duty, 400 mA/cm², 500 cm² electrode area [10].

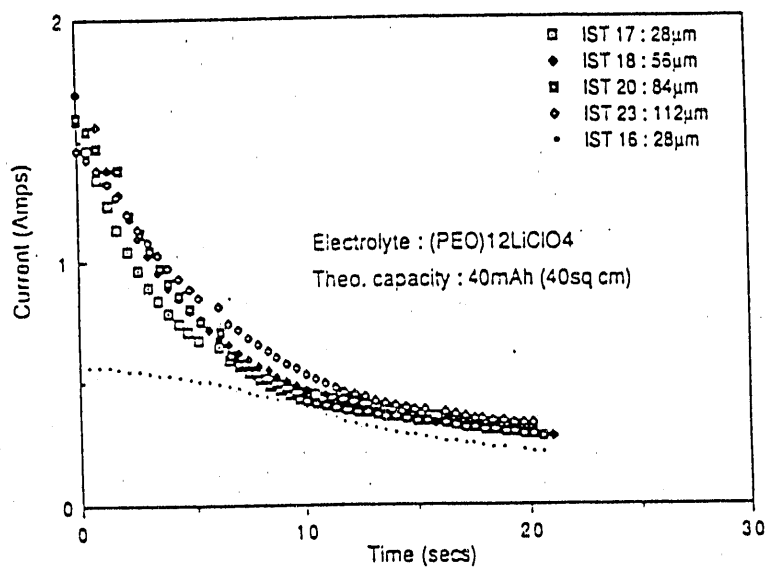


Figure 7. Typical first pulses of several Li/poly(ethylene-oxide)₁₂LiClO₄/V₆O₁₃ cells in 20-s long 1.7-V potentiostatic pulses at 120°C [13]. Cells were built with electrolyte films of the indicated thicknesses.

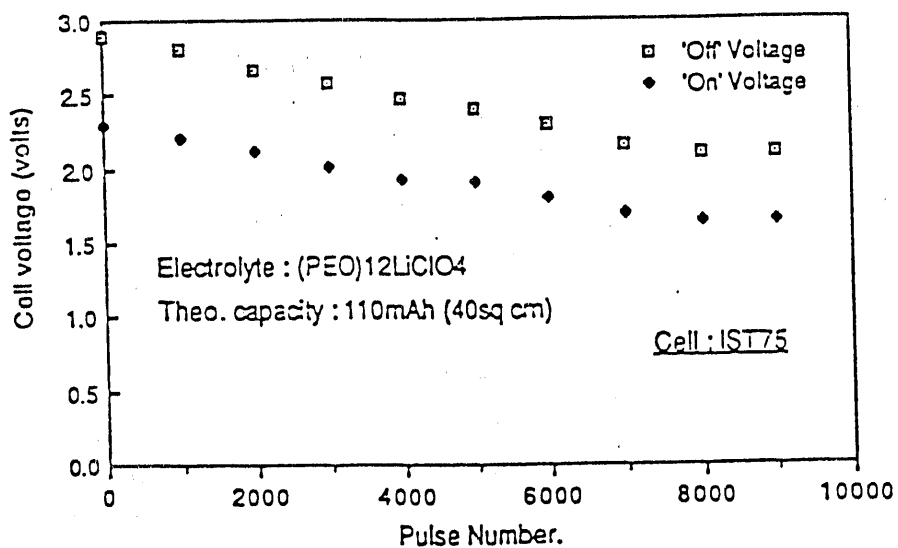


Figure 8. Cell voltage in repeated 10-ms long pulses at 120°C of a Li/poly(ethylene-oxide)₁₂LiClO₄/V₆O₁₃ cell; average current: 1.3 A, cumulative capacity: 25 mAh [13].

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KEY WORDS

Page 1. bipolar batteries, pulse batteries, lithium batteries

Page 2. ultra capacitor, galvanic capacitor

Page 3. Area-specific impedance, thermal management

Page 4. Thermoneutral operation, disulfide batteries

Page 5. Li-alloy/chlorine battery, Li/polymer electrolyte battery

Page 6. Li/SOCl₂ battery, electric vehicle, battery-battery hybrid.

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